

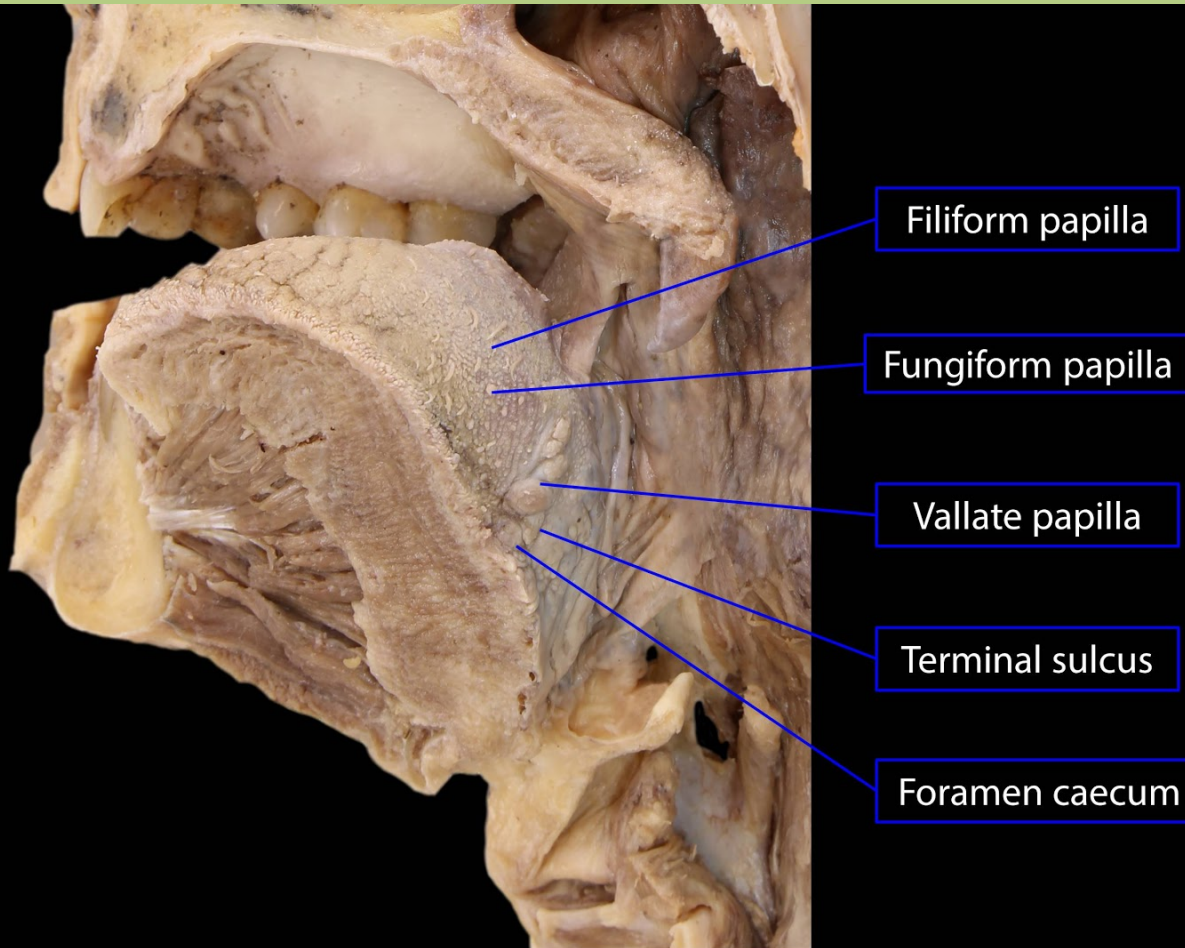
Histology of the tongue

Covering epithelium:

- *stratified non-keratinized squamous epithelium*

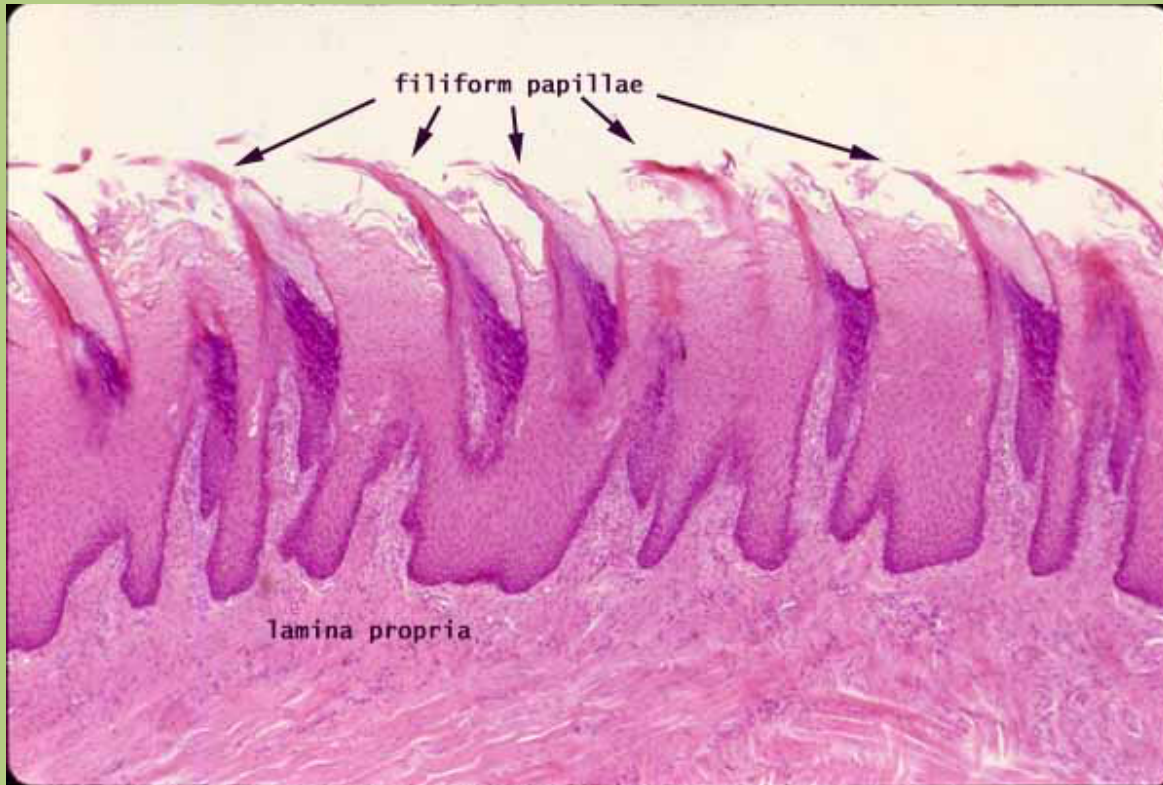
Papillae:

- *filiform*
- *fungiform*
- *foliate*
- *circumvallate*



External tongue, posteromedial

Histology of the tongue



Filiform papillae:

- *conical, elongated projections*
- *covered by the highly keratinized stratified epithelium*
- *do not represent taste buds*
- *they have mechanical role*

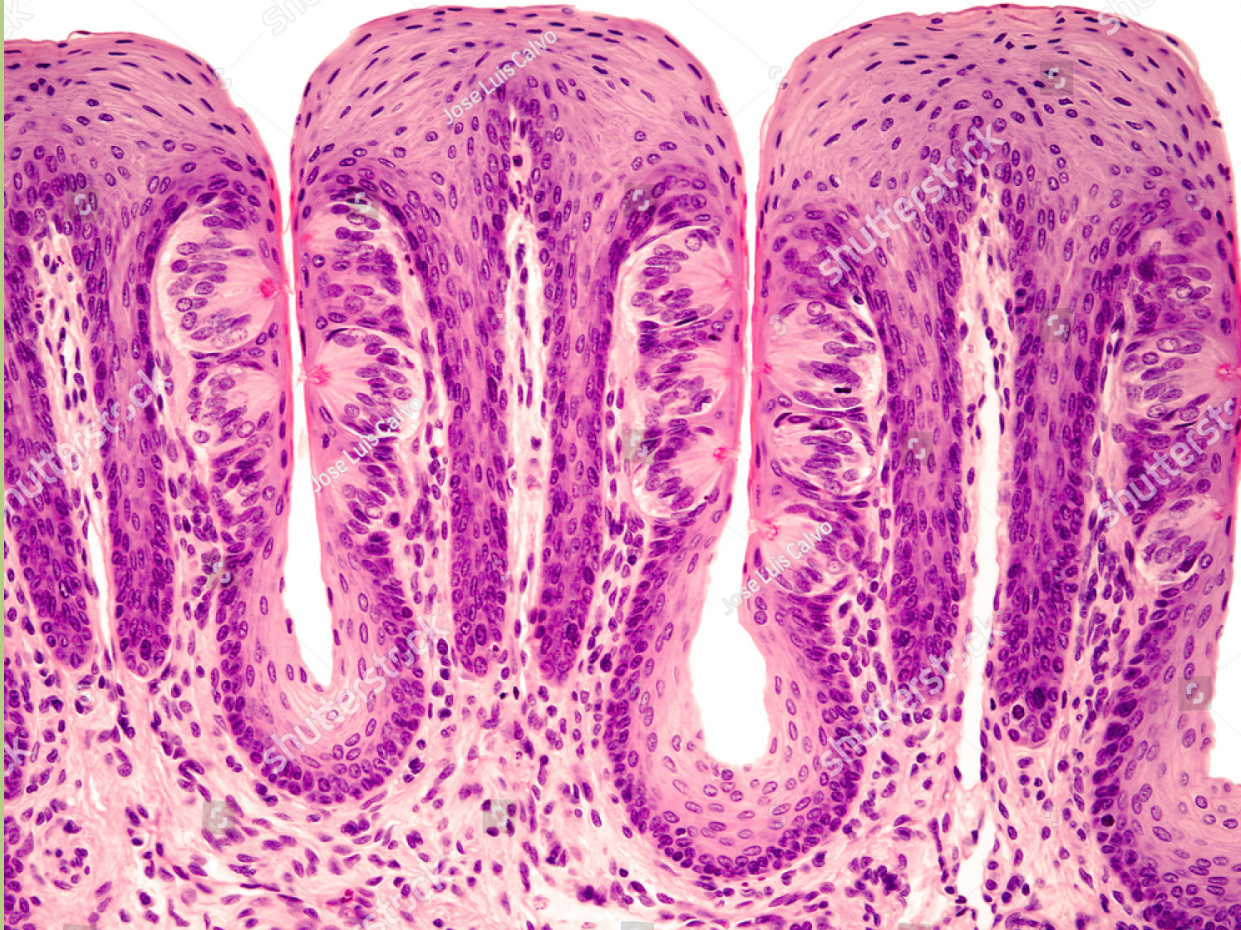
Histology of the tongue

Fungiform papillae:

- *mushroom-shaped projections*
- *taste buds are presented in the stratified squamous epithelium on the dorsal surface of papillae*



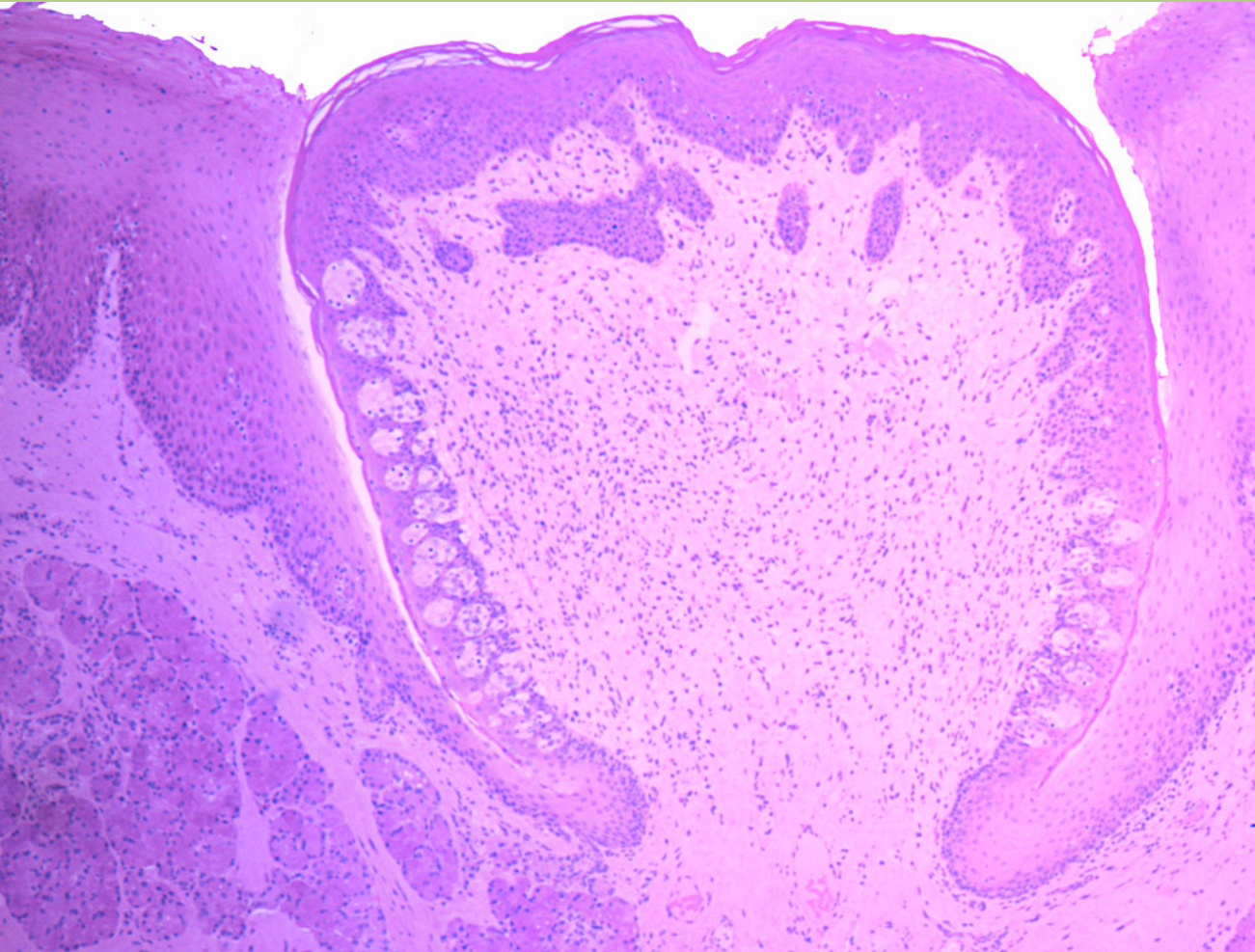
Histology of the tongue



Foliate papillae:

- *consist of parallel low ridges separated by deep mucosal clefts*
- *taste buds are presented on the lateral sides*
- *small serous glands empty into the the clefts*

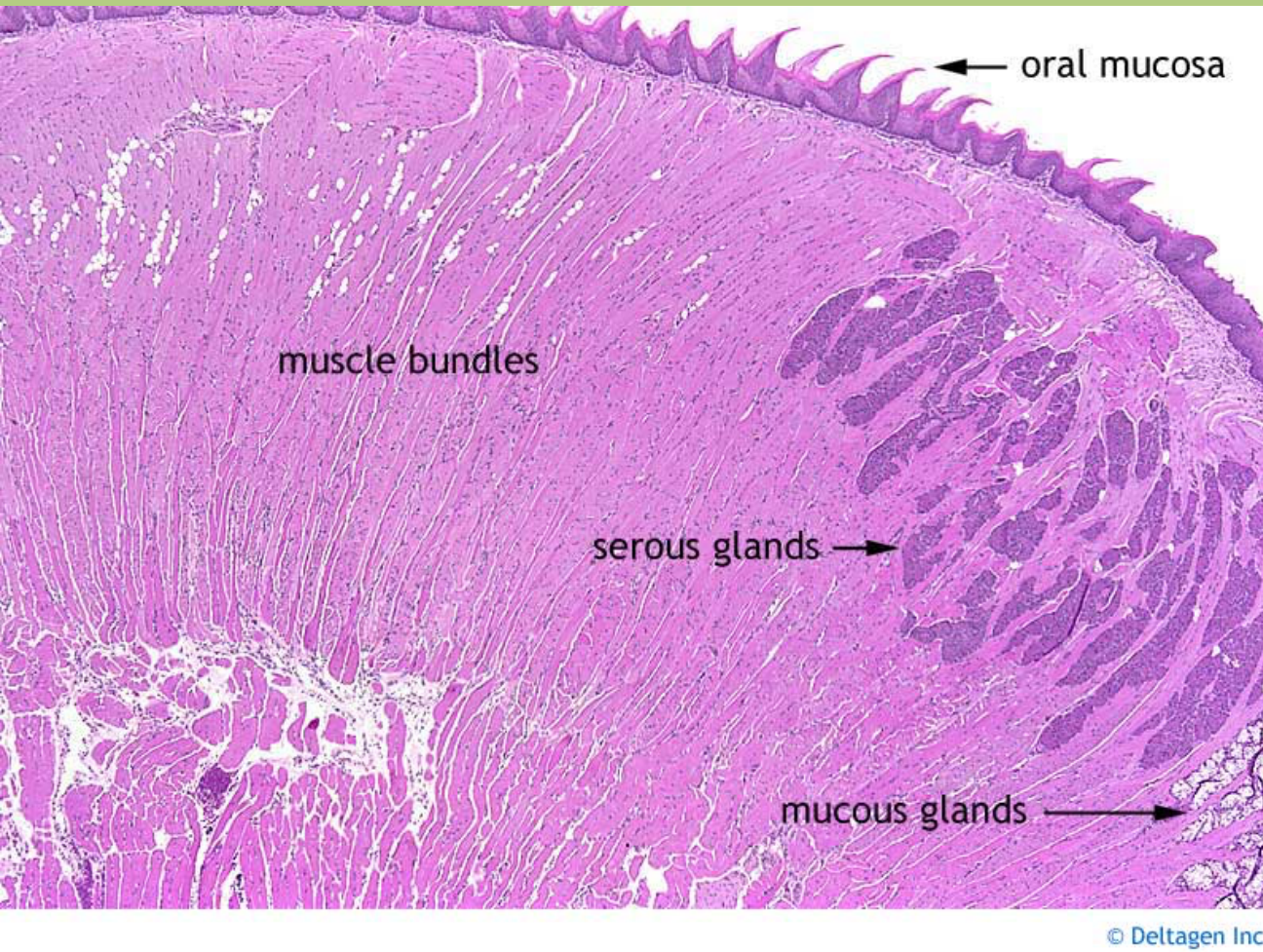
Histology of the tongue



Circumvallate papillae:

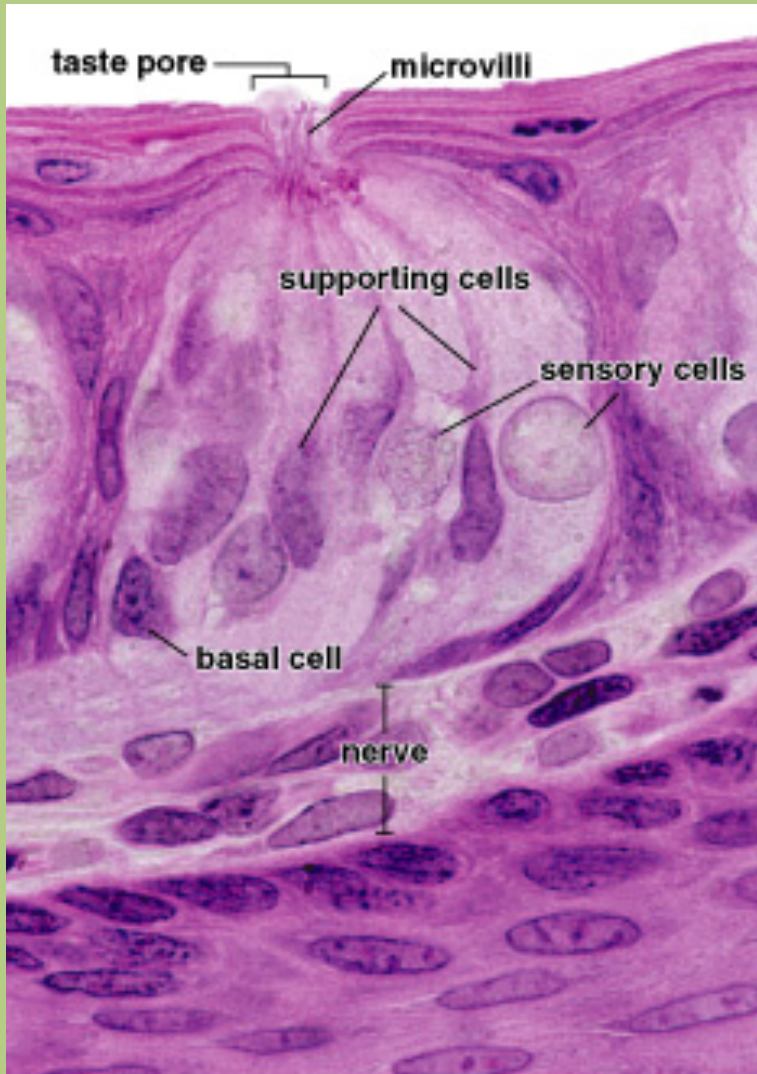
- *8 to 12 papillae locate in front of the terminal sulcus*
- *taste buds are presented on the lateral sides*
- *each papilla surrounded by a moatlike invagination lined with stratified squamous epithelium*
- *von Ebner's glands empty their serous secretion into the base of the moatlike invaginations*

Histology of the tongue



- *glands*
- *striated visceral muscle fibers*
- *stratified non-keratinized squamous epithelium*

Taste buds



Sensory cells:

- the most numerous cells
- they show microvilli
- at their base they form a synapse with the processes of afferent sensory neurons
- turnover time is about 10 days

Supporting cells:

- they represent microvilli and tight junctions
- turnover time is about 10 days

Basal cells:

- small cells located in the basal portion, near to the basal lamina
- they are the stem cells for the two other cell types